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Diagnostic accuracy of a machine learning model using radiomics features from breast synthetic MRI

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Abstract

Background In breast magnetic resonance imaging (MRI), the differentiation between benign and malignant breast masses relies on the Breast Imaging Reporting and Data System Magnetic Resonance Imaging (BI-RADS-MRI) lexicon. While BI-RADS-MRI classification demonstrates high sensitivity, specificities vary. This study aimed to evaluate the feasibility of machine learning models utilizing radiomics features derived from synthetic MRI to distinguish benign from malignant breast masses.

Methods Patients who underwent breast MRI, including a multi-dynamic multi-echo (MDME) sequence using 3.0 T MRI, and had histopathologically diagnosed enhanced breast mass lesions were retrospectively included. Clinical features, lesion shape features, texture features, and textural evaluation metrics were extracted. Machine learning models were trained and evaluated, and an ensemble model integrating BI-RADS and the machine learning model was also assessed.

Results A total of 199 lesions (48 benign, 151 malignant) in 199 patients were included in the cross-validation dataset, while 43 lesions (15 benign, 28 malignant) in 40 new patients were included in the test dataset. For the test dataset, the sensitivity, specificity, accuracy, and area under the curve (AUC) of the receiver operating characteristic for BI-RADS were 100%, 33.3%, 76.7%, and 0.667, respectively. The logistic regression model yielded 64.3% sensitivity, 80.0% specificity, 69.8% accuracy, and an AUC of 0.707. The ensemble model achieved 82.1% sensitivity, 86.7% specificity, 83.7% accuracy, and an AUC of 0.883. The AUC of the ensemble model was significantly larger than that of both BI-RADS and the machine learning model.

Conclusion The ensemble model integrating BI-RADS and machine learning improved lesion classification.

Keywords Breast MRI, Breast cancer, Artificial intelligence, Machine learning, Binary classification

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Background

Breast cancer is the most common cancer among women and the fourth leading cause of cancer-related death among women in Japan [1]. In the precise evaluation of breast lesions, the differentiation between benign and malignant lesions is often based on category classification system in the Breast Imaging Reporting And Data System Magnetic Resonance Imaging (BI-RADS-MRI) lexicon [2, 3]. If a lesion is classified as BI-RADS category 4 or 5 and considered potentially malignant, a histopathological examination such as a needle biopsy, is typically performed.

A well-documented limitation of BI-RADS-MRI is the subjectivity of its criteria, particularly in category 4, which encompasses lesions with a broad range of malignant potential (>2% but <95%) [2]. BI-RADS-MRI classification demonstrates high sensitivity (94–100%), although reported specificity varies widely (37–97%) [4]. As a result, despite malignancy being suspected on dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI), histopathological confirmation frequently reveals benign findings. The high false-positive rate of BI-RADS-MRI leads to unnecessary biopsies, contributing to increased procedural invasiveness and healthcare costs.

Quantitative value maps, such as T1 and T2 relaxation times, can be obtained using MRI. Previous studies have reported that these quantitative values aid in differentiating benign from malignant breast lesions [5, 6]. These quantitative value maps have been considered promising

diagnostic tools for breast imaging. However, each value requires a dedicated scan sequence, as in conventional MRI sequences. Their clinical application has been challenging due to extended scan times, which limit the feasibility of routine breast MRI protocols.

“SyMRI” (SyntheticMR, Linköping, Sweden) which is one of synthetic MRI techniques enable simultaneous acquisition of quantitative T1, T2, and proton density (PD) maps from a single sequence. In this application, images are acquired using a two-dimensional fast spin echo (2D FSE) multi-dynamic, multi-echo (MDME) sequence that collects eight complex-valued images per slice with four saturation delays and two echo times (Fig. 1) [7]. As reported in some previous studies, these quantitative values obtained from synthetic MRI demonstrate good diagnostic performance in differentiating benign from malignant breast lesions [8, 9].

Radiomics is an advanced imaging analysis method that extracts numerous features, including morphology, histogram, and texture characteristics, from clinical images, enhancing diagnostic accuracy by identifying key imaging biomarkers [10]. Radiomics features derived from breast MRI can distinguish between benign and malignant lesions [11, 12] and predict the response to neoadjuvant chemotherapy (NAC) [13–15]. However, the MRI scans used in these studies are relative value images rather than quantitative value maps, meaning that radiomics feature extraction may be influenced by differences in imaging protocols, institutions, and MRI vendors.

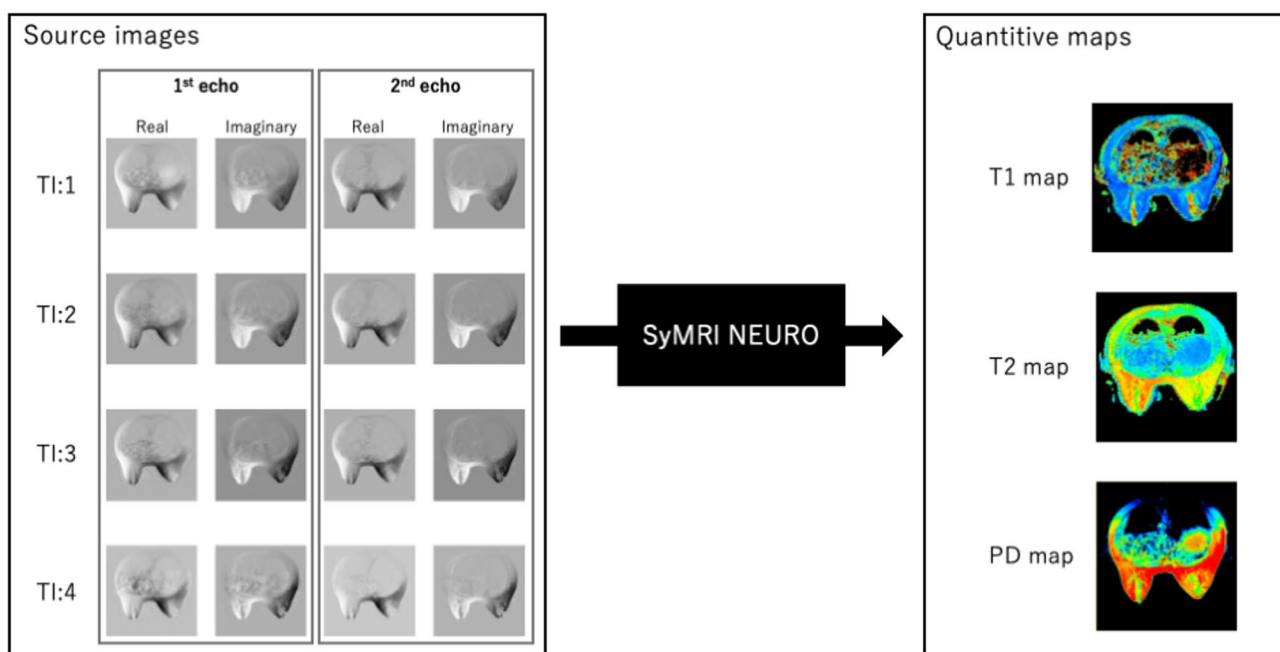


Fig. 1 Schematic overview of synthetic MRI. Complex-valued images were acquired using four saturation delays and two echo times. Post-processing was performed using the dedicated commercial application SyMRI NEURO to generate quantitative T1, T2, and proton density (PD) maps

Hwang et al. evaluated the ability of a logistic regression model incorporating radiomics features derived from synthetic MRI to predict pathologic complete response to NAC in 181 participants with triple-negative breast cancer (TNBC). Their findings suggested that quantitative maps obtained from synthetic MRI-based radiomics features, particularly when acquired at mid-treatment, may be useful for identifying early NAC responders in TNBC [16]. However, to the best of our knowledge, no other studies have systematically examined the clinical utility of synthetic MRI-derived quantitative maps for radiomics-based or artificial intelligence (AI)-driven breast imaging applications.

The purpose of this study was to investigate the diagnostic accuracy of machine learning models incorporating radiomics features extracted from synthetic MRI images of the breast, both before and after contrast enhancement, to differentiate between benign and malignant breast mass lesions.

Methods

Clinical data

This study was approved by the institutional review board of the author's hospital. Patients who underwent contrast-enhanced breast MRI with a protocol including an MDME sequence on a 3.0 T MRI scanner (SIGNA Architect, GE HealthCare, Milwaukee, WI) between February 2018 and August 2022 were retrospectively included. A histopathological diagnosis obtained via biopsy or surgery was considered the gold standard for lesion classification as benign or malignant.

The inclusion criteria consisted of patients who underwent breast MRI with pre- and post-contrast MDME sequences. Patients were excluded if (1) no breast lesion in breast MRI was identified, (2) only non-mass lesions were present, (3) the lesion was too small to be detected on quantitative value maps, (4) the histopathological diagnosis was inconclusive regarding malignancy, or (5) NAC had been administered before MRI.

Patients were divided into training and test cohorts. The test cohort consisted of a temporally independent consecutive case series. In the training cohort, if multiple masses were present in a single patient, only the largest

mass per patient was included. In the test cohort, if multiple masses were present in a single case, the largest mass per breast was included.

MRI imaging protocols and quantitative image analysis

Breast MRI was performed using a 3.0 T system (SIGNA Architect, GE HealthCare, Milwaukee, WI) with an eight-channel phased-array breast surface coil. All patients were scanned in the prone position. The imaging sequences included in this study are shown in Fig. 2. Axial MDME sequences were acquired before and after contrast injection as part of this protocol. Synthetic MRI data were obtained using MDME sequences, which were performed with an interleaved slice-selective 120-degree saturation and multi-echo acquisition [17, 18]. The MDME sequence incorporated two echo times and four automatically calculated saturation delays, generating eight real and eight imaginary images per acquisition for quantification of longitudinal (T1) and transverse (T2) relaxation times. Quantitative T1 and T2 relaxation time maps were calculated using SyMRI NEURO software (Synthetic MR, Linköping, Sweden). The total scan time for the MDME sequence was 3.53 min. In this study, MDME sequences were acquired before and after the DCE-MRI sequence. The post-contrast MDME sequence was initiated 12 min after contrast injection. The MDME sequence parameters were as follows: repetition time (TR), 4470 ms; echo time (TE), 15.5/93.1 ms; field of view (FOV), 360 × 360 mm; matrix size, 256 × 224; slice thickness, 4 mm; interslice gap, 0.4 mm; number of slices, 28; echo-train length, 16; and acceleration factor, 2.5. Fat-saturation T1-weighted DCE-MRI was acquired following the administration of a gadolinium-based contrast agent. The DCE-MRI sequence included one pre-contrast and multiple post-contrast acquisitions.

The DCE-MRI acquisitions were performed using a T1-weighted 3D fast spoiled gradient-recalled echo sequence with parallel volume imaging for breast assessment (Volume Imaged Breast Assessment [VIBRANT], GE HealthCare). The DCE-MRI sequence parameters were as follows: TR, 7.2 ms; TE, 2.4 ms; flip angle, 10°; FOV, 350 × 329 mm; number of slices, 224; slice

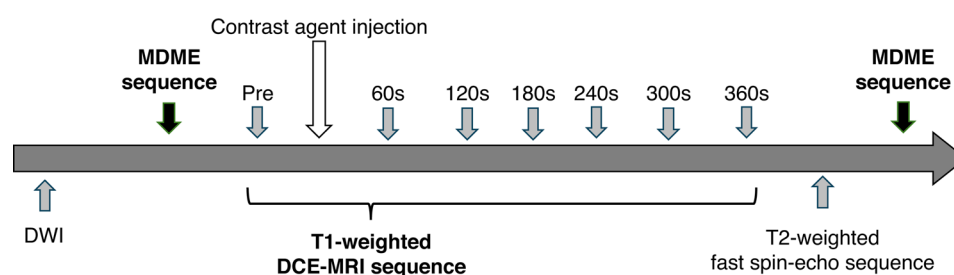


Fig. 2 The MRI protocol of this study. MDME; multi-dynamic multi-echo, DWI: diffusion-weighted imaging, DCE-MRI: dynamic contrast-enhanced MRI

thickness, 1.4 mm; and acquisition matrix, 350×256 . Following an intravenous injection of a gadolinium-based contrast agent at a rate of 2 mL/s and a dose of 0.2 mmol/kg, six post-contrast dynamic scans were acquired. The first post-contrast sequence was acquired 60 s after contrast injection, with the subsequent five series acquired at 60-s intervals.

Quantitative T1 and T2 relaxation time maps (Q-maps) were obtained from MDME sequence images. A T1/T2 ratio map was created by calculating the pixel-wise ratio of T1 and T2 values. As both T1 and T2 maps were generated from a single MDME pulse sequence, no misregistration occurred between them, enabling precise pixel-based ratio calculations.

Lesion segmentation on synthetic MRI

The local extent of lesion boundaries on synthetic MRI was unclear, making precise annotation challenging (Additional file 1). In general, the boundaries of mass lesions are clearly enhanced in DCE-MRI. Therefore, high-resolution DCE-MRI acquired 120 s after contrast injection was used for segmentation. To ensure that the number of slices covering the entire lesion remained consistent across imaging datasets, DCE-MRI images were resliced to match the slice thickness of the quantitative map images.

A board-certified radiologist identified the lesion on all datasets using the resliced DCE-MRI images and manually delineated the visible lesion boundary on all slices containing the lesion using 3D segmentation tools in Insight Toolkit-SNAP (ITK-SNAP) software [19]. The delineated lesion boundaries on DCE-MRI were transformed to the pre- and post-contrast Q-map images by the method detailed in the following “Image Registration” section. To evaluate interobserver variability in lesion boundary annotation, 30 randomly selected lesions were independently segmented by a second radiologist, who was blinded to the original annotations. The degree of agreement between the two radiologists was assessed using the Dice similarity coefficient. Additionally, the Dice coefficient for lesion boundaries incorporating inflation-based adjustments was also calculated.

Image registration

As the scans were performed at two different time points (before and after contrast injection), patient motion may have occurred. This motion resulted in misregistration between DCE-MRI and MDME images. To correct this misalignment, 3D co-registration was performed between the DCE-MRI and MDME image volumes (Additional file 2). Because the breast consists of soft tissue that is susceptible to deformation, misregistration was primarily caused by non-linear local deformations of the breast. To account for this localized deformation,

the affected breast was automatically cropped from the first-echo MDME magnitude image and DCE-MRI. A 3D symmetric normalization approach (Affine + deformable transformation, using mutual information as the optimization metric) was applied to calculate transformation parameters using the open-source application ANTsPy [20]. In this process, the MDME image was designated as the fixed reference, while the DCE-MRI image was treated as the moving dataset. The computed registration parameters were then used to transform the lesion boundary delineated on DCE-MRI to the pre- and post-contrast Q-map images.

Feature extraction and analysis

The feature extraction process in this study is summarized in Fig. 3. Clinical features were obtained from electronic medical records, while lesion-shape features were extracted from the region of interest on DCE-MRI. Additionally, textural evaluation metrics were computed from texture features derived from the T1/T2 ratio maps.

Clinical features

Clinical features included age, menopausal status, and presence or absence of symptoms, extracted from electronic medical records at the time of breast MRI.

Lesion shape features

Fourteen morphological shape features were extracted from the lesion boundary on DCE-MRI, as annotated by a radiologist. The boundary was smoothed and inflated using binary dilation with a 3×3 kernel mask.

Radiomics features extraction of T1/T2

A radiomics feature mapping framework [21] transformed each 3D breast lesion on T1/T2 ratio maps into a multidimensional radiomics feature space (RFS), defined as $RFS_{T1/T2} = \{RF_1, RF_2 \dots RF_n\} \in R^D$, where RF_i represents the i -th mean radiomics feature value calculated on all lesion voxels, D represents the number of voxels in the lesion, and n represents the number of RFs generated, which is 93. Radiomics features were categorized into first-order and second-order texture descriptors. First-order features quantify intensity distribution within the lesion, while second-order features capture spatial relationships between voxels. Table 1 provides details of the extracted features.

Textural evaluation metric for T1/T2 map image

The textural evaluation metric (TEM) [11] was used to assess changes in tissue textural feature vectors before and after contrast injection. In this study, $RFS_{T1/T2}$ of pre- and post-contrast were used to calculate $TEM_{T1/T2}$.

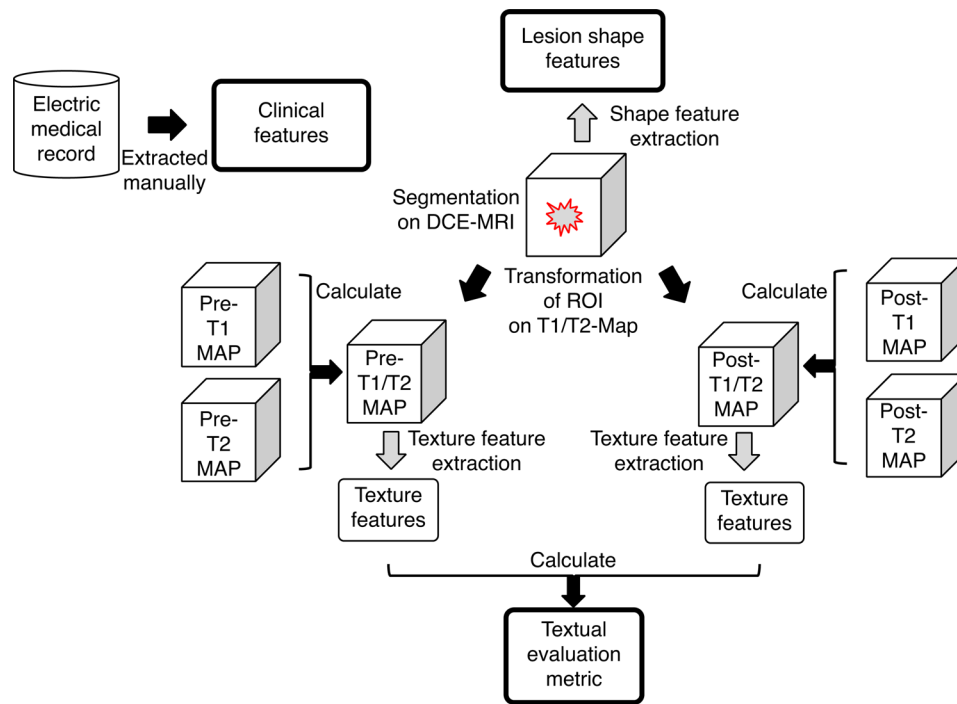


Fig. 3 Summary of the feature extraction process. Clinical features, lesion shape features, and textural evaluation metrics were extracted from electronic medical records and MRI images

Table 1 Radiomics features extracted in the study

Feature type	Feature family	No. features
Shape	Morphology	14
First-order Texture	Local Intensity, Intensity-based Statistics, Intensity Histogram	18
Second-order Texture	Grey Level Co-occurrence Matrix (GLCM)	24
	Gray Level Dependence Matrix (GLDM)	14
	Grey Level Run Length Matrix (GLRLM)	16
	Grey Level Size Zone Matrix (GLSZM)	16
	Neighborhood Grey Tone Difference Matrix (NGTDM)	5

Total: 107 (14 shape + 93 texture features)

TEM_{T_1/T_2} consists of 93 textural feature change values per lesion and is defined below.

$$TEM_{T_1/T_2} = \frac{RFS_{T_1/T_2 \text{ pre}}}{RFS_{T_1/T_2 \text{ post}}}$$

Feature selection

Clinical features, shape features, and TEM_{T_1/T_2} were used as input variables. Feature selection aimed to retain the most reliable and relevant features while removing redundant information, improving model robustness, reducing overfitting risk, and optimizing computation time. For each fold in the training dataset, features were selected using the Boruta algorithm [22], and

additional features were removed to ensure the Variance Inflation Factor (VIF) remained ≤ 500 to minimize multicollinearity.

Model building

This study evaluated multiple machine learning models, including random forest (RF), extreme gradient boosting (XGB), gradient boosting (GB), light gradient boosting (LGB), logistic regression (LR), k-nearest neighbor (KNN), classification and regression tree (CART), support vector machine (SVM), Gaussian naïve Bayes, and a stacked model combining LR, KNN, CART, SVM, and Bayes classifiers. A stratified five-fold cross-validation was performed on the training dataset. Due to class imbalance, where benign lesions were the minority class, the Adaptive Synthetic Sampling Approach for Imbalanced Learning (ADASYN) algorithm [23] was applied in each fold to oversample benign lesions to match the malignant class. All models were trained on selected meaningful features and validated to generate performance metrics. Diagnostic performance was assessed using mean sensitivity, specificity, accuracy, and the area under the curve (AUC) of the receiver operating characteristic (ROC) for malignancy classification. The best-performing model was determined by the highest mean AUC across the five-fold validation.

The final model performance was evaluated on the test dataset after training on the full training set. Additionally, feature importance was analyzed using SHapley

Additive exPlanations (SHAP) values [24]. SHAP, based on cooperative game theory, quantifies each feature's contribution to the model's prediction without assessing prediction quality.

BI-RADS-MRI

Breast lesion categorization in the test cohort was performed using the BI-RADS-MRI lexicon from BI-RADS 5th. Two radiologists independently evaluated contrast-enhanced MR images, and in cases of disagreement, a final category was determined by consensus. For the BI-RADS-MRI assessment, category 3 masses were defined as benign, and category 4 or 5 masses were defined as malignant. Sensitivity, specificity, accuracy, and AUC were calculated using histopathological diagnosis as the gold standard.

Ensemble algorithm

An ensemble algorithm integrating BI-RADS-MRI and the trained machine learning model was applied. Category 3 lesions remained benign, and category 5 lesions remained malignant, while category 4 lesions were reclassified by the machine learning model. Performance metrics, including sensitivity, specificity, accuracy, and AUC, were calculated based on histopathological diagnosis as the reference standard.

Statistical analysis

The mean Dice coefficient with and without inflation was compared using the Wilcoxon rank-sum test. BI-RADS-MRI category agreement between two radiologists was assessed using the weighted Cohen's Kappa coefficient. The AUCs of BI-RADS-MRI, the machine learning model, and the ensemble method were compared using the DeLong test with Bonferroni correction (three comparisons). A significance level of $p < 0.05$ was applied. Statistical analyses were conducted using R software (version 4.4.1; R Foundation for Statistical Computing).

Computer environment

Machine learning was conducted on Ubuntu 20.04 Long-Term Support using an Intel Xeon Silver 4108 CPU (8 cores, 1.80 GHz) and an NVIDIA Quadro Ray Tracing Texel eXtreme (RTX) 5000 Graphics Processing Unit.

Results

Dataset split

A total of 471 patients who underwent breast MRI with pre- and post-contrast MDME sequences were included. Patients were excluded based on the following criteria: 44 patients in whom no enhanced lesion in breast MRI was detected, 92 patients with non-mass lesions, 12 patients with small lesions, 33 patients with unclear benign or malignant status in histopathological diagnosis, and 51

patients who received NAC before MRI. After excluding 232 patients, the final dataset comprised 199 lesions (48 benign, 151 malignant) from 199 patients for cross-validation and 43 lesions (15 benign, 28 malignant) from 40 patients for testing. Subtype details are provided in Additional files 3 and 4.

Inter-annotator agreement

The mean Dice coefficients for the two radiologists were 0.845 ± 0.086 (95% Confidence Interval (CI), 0.813–0.878) with inflation and 0.783 ± 0.116 (95% CI, 0.740–0.827) without inflation. Dice scores with inflation were significantly higher than those without inflation ($p < 0.001$).

BI-RADS-MRI

The weighted Cohen's Kappa coefficient for BI-RADS category agreement between the two radiologists was 0.853 (95% CI, 0.796–0.911), indicating strong interobserver agreement.

In the final BI-RADS assessment categories, the cross-validation dataset contained 17 lesions in category 3 (17 benign), 119 in category 4 (30 benign, 89 malignant), and 63 in category 5 (1 benign, 62 malignant). For the cross-validation dataset, BI-RADS achieved sensitivity, specificity, accuracy, and AUC value of 100% (95% CI, 97.6–100.0), 35.4% (95% CI, 22.2–50.5), 84.4% (95% CI, 78.6–89.2), and 0.677 (95% CI, 0.609–0.745), respectively. Accuracy by category was 100% for category 3, 74.8% (95% CI, 66.0–82.3) for category 4, and 98.4% (95% CI, 91.5–100.0) for category 5.

The test dataset contained 5 lesions in category 3 (5 benign), 29 in category 4 (10 benign, 19 malignant), and 9 in category 5 (9 malignant). For the test dataset, BI-RADS achieved sensitivity, specificity, accuracy, and AUC value of 100% (95% CI, 87.6–100.0), 33.3% (95% CI, 11.8–61.6), 76.7% (95% CI, 61.4–88.2), and 0.667 (95% CI, 0.543–0.790), respectively. Accuracy was 100% for categories 3 and 5, and 65.5% (95% CI, 45.7–82.0) for category 4.

Machine learning models

The average diagnostic performance of each machine learning model on the stratified five-fold cross-validation dataset is shown in Table 2. The model with the highest AUC was LR, with the average $\pm$ standard deviation of the sensitivity, specificity, accuracy, and AUC value of $65.5 \pm 9.6\%$, $79.3 \pm 14.7\%$, $68.8 \pm 8.0\%$, and 0.724 ± 0.087 , respectively.

On the test dataset, LR achieved sensitivity, specificity, accuracy, and AUC value of 64.3% (95% CI, 44.1–81.4), 80.0% (95% CI, 51.9–95.7), 69.8% (95% CI, 53.9–82.8), and 0.707 (95% CI, 0.535–0.879), respectively. On the subset of BI-RADS category 3 (all benign), LR achieved accuracy of 80.0% (95% CI, 28.4–99.5). On the subset of BI-RADS category 4, LR achieved sensitivity, specificity,

Table 2 Average diagnostic performance of machine learning models in stratified five-fold cross-validation

Model	Mean sensitivity	Mean specificity	Mean accuracy	Mean AUC
RF	0.715 ± 0.062	0.605 ± 0.141	0.687 ± 0.046	0.660 ± 0.064
XGB	0.781 ± 0.092	0.502 ± 0.127	0.713 ± 0.046	0.642 ± 0.034
GB	0.772 ± 0.087	0.610 ± 0.134	0.732 ± 0.079	0.691 ± 0.086
LGB	0.782 ± 0.068	0.524 ± 0.183	0.719 ± 0.053	0.653 ± 0.084
LR	0.655 ± 0.096	0.793 ± 0.147	0.688 ± 0.080	0.724 ± 0.087
KNN	0.655 ± 0.073	0.604 ± 0.152	0.643 ± 0.021	0.630 ± 0.041
CART	0.730 ± 0.077	0.539 ± 0.094	0.683 ± 0.043	0.635 ± 0.031
SVM	0.689 ± 0.069	0.629 ± 0.092	0.673 ± 0.049	0.659 ± 0.048
Bayes	0.675 ± 0.110	0.689 ± 0.165	0.679 ± 0.119	0.682 ± 0.132
Stacking	0.731 ± 0.075	0.541 ± 0.095	0.684 ± 0.042	0.636 ± 0.031

All values are expressed as mean ± standard deviation

accuracy, and AUC value of 73.7% (95% CI, 48.8–90.9), 80.0% (95% CI, 44.4–97.5), 75.9% (95% CI, 56.6–89.7), and 0.695 (95% CI, 0.477–0.912), respectively. On the subset of BI-RADS category 5 (all malignant), LR achieved accuracy of 44.4% (95% CI, 13.7–78.8).

The importance and impact of eight selected features were analyzed, and results are shown in two summary plots (Fig. 4). A horizontal bar plot ranks feature effects from highest to lowest, while a beeswarm violin plot visualizes their impact on benign and malignant lesion predictions. In this analysis, menopausal status ranked highest, while patient age was seventh. A single shape feature (minor axis length) ranked sixth, while the remaining five features were from TEM_{T_1/T_2} .

Ensemble algorithm performance

The ensemble algorithm, combining BI-RADS and the trained LR model, achieved sensitivity, specificity, accuracy, and AUC value of 82.1% (95% CI, 63.1–93.9), 86.7%

(95% CI, 59.5–98.3), 83.7% (95% CI, 69.3–93.2), and 0.883 (95% CI, 0.773–0.994) on the test dataset, respectively. The performance on the test dataset of BI-RADS, the machine learning model, and ensemble were summarized in Table 3.

The ROC curves for the test dataset, comparing BI-RADS, the machine learning model, and the ensemble method, are shown in Fig. 5. The AUC values were 0.667 (95% CI, 0.543–0.790) for BI-RADS, 0.707 (95% CI, 0.535–0.879) for the machine learning model, and 0.883 (95% CI, 0.773–0.994) for the ensemble method. No significant difference was found between BI-RADS and the machine learning model ($p=0.699$). However, the AUC of the ensemble algorithm was significantly higher than that of BI-RADS ($p=0.001057$) and the machine learning model ($p=0.005074$). Applying Bonferroni correction for multiple comparisons (three tests), the statistical significance threshold was set at $p<0.0167$.

Discussion

The LR model demonstrated the highest diagnostic performance among the machine learning models. Key predictive features included menopausal status, minor axis length, age, and texture feature ratios. The ensemble algorithm outperformed individual models, achieving a significantly higher AUC than those of both BI-RADS and the machine learning model.

MRI scan protocols acquire images in multiple independent sequences and planes. T1- and T2-weighted images are visually assessed for signal intensity, but signal intensities from relative images cannot be directly compared across examinations or patients due to variations in acquisition parameters and scanner settings. Moreover, image contrast in these sequences is influenced not only by tissue relaxation times but also by radio

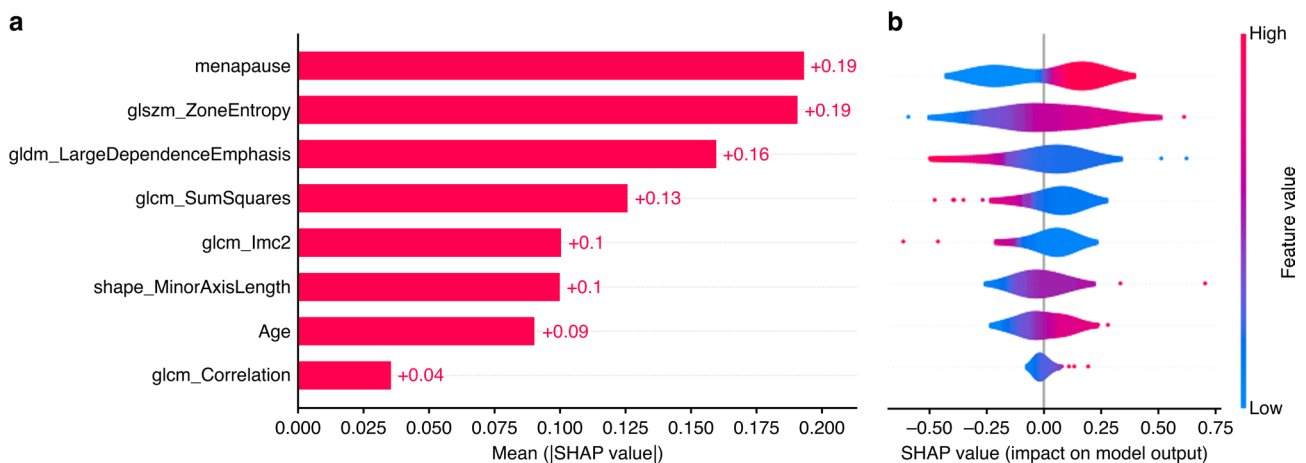


Fig. 4 Feature importance in our proposed model using SHAP analysis. **(a)** A horizontal bar plot ranks feature effects from highest to lowest. **(b)** A beeswarm violin plot visualizes how high and low feature values impact prediction outcomes

Table 3 Overall and category-stratified performance on the test dataset

Model	BI-RADS subset (n; benign/malignant)	Sensitivity, % (95% CI)	Specificity, % (95% CI)	Accuracy, % (95% CI)	AUC (95% CI)
BI-RADS	All (43; 15/28)	100.0 (87.6–100.0)	33.3 (11.8–61.6)	76.7 (61.4–88.2)	0.667 (0.543–0.790)
	3 (5; 5/0)	-	100.0 (47.8–100.0)	100.0 (47.8–100.0)	-
	4 (29; 10/19)	100.0 (82.4–100.0)	-	65.5 (45.7–82.0)	-
	5 (9; 0/9)	100.0 (66.4–100.0)	-	100.0 (66.4–100.0)	-
Machine Learning	All (43; 15/28)	64.3 (44.1–81.4)	80.0 (51.9–95.7)	69.8 (53.9–82.8)	0.707 (0.535–0.879)
	3 (5; 5/0)	-	80.0 (28.4–99.5)	80.0 (28.4–99.5)	-
	4 (29; 10/19)	73.7 (48.8–90.9)	80.0 (44.4–97.5)	75.9 (56.6–89.7)	0.695 (0.477–0.912)
	5 (9; 0/9)	44.4 (13.7–78.8)	-	44.4 (13.7–78.8)	-
Ensemble	All (43; 15/28)	82.1 (63.1–93.9)	86.7 (59.5–98.3)	83.7 (69.3–93.2)	0.883 (0.773–0.994)

The bold text means the overall performance of test dataset

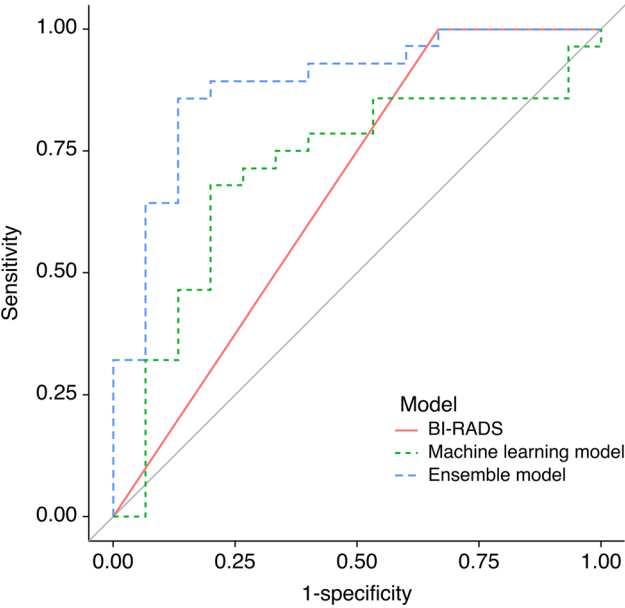


Fig. 5 ROC curve for predicting malignant cases in the test dataset. The AUC values were 0.667 for BI-RADS, 0.707 for the machine learning model, and 0.862 for the ensemble method. The ensemble model's AUC was significantly higher than that of BI-RADS ($p=0.001057$) and the machine learning model ($p=0.005074$). With Bonferroni correction for three comparisons, the significance threshold was set at $p < 0.0167$

frequency field variations, coil sensitivity, proton density, and other factors [25, 26]. These variations complicate accurate lesion signal evaluation. In particular, changes in signal intensity before and after contrast injection lack a quantitative standard, potentially reducing interobserver consistency and diagnostic accuracy.

Quantitative MRI values, such as T1 and T2, have emerged as promising tools for lesion characterization. Previous studies suggest these values aid in differentiating benign from malignant breast lesions [5, 6]. However, obtaining separate quantitative value maps for each sequence extends examination time. Synthetic MRI requires only an MDME sequence, which takes approximately 5 min and generates multiple quantitative value maps through post-processing. In this study, we used synthetic MRI to obtain quantitative value maps of the breast before and after contrast enhancement.

T1/T2 mapping has been investigated for differentiating benign from malignant lesions in various organs, including the breast [27, 28]. In breast lesion, Medina et al. demonstrated that the T1/T2 ratio significantly improved discrimination between fibroadenomas and adenocarcinomas ($p < 0.05$) [29]. Similarly, Malikova et al. reported a statistically significant difference in T1/T2 ratios between benign and malignant breast lesions [27]. As both T1 and T2 maps are generated from a single MDME pulse sequence, no misregistration occurs, enabling accurate pixel-based ratio calculations. Radiomics features were extracted from pre- and post-contrast T1/T2 map images.

Previous studies indicate that visual support from additional images (e.g., post-contrast T1-weighted images) is necessary for radiomics feature extraction, as lesion contours are often unclear in quantitative value maps [16]. In this study, lesion segmentation information from DCE-MRI was deformed using deformable image registration and transcribed to quantitative value maps before and after contrast enhancement. This approach facilitated segmentation and improved radiomics feature extraction.

BI-RADS-MRI categorization is widely used for assessing benign and malignant breast masses in clinical settings [2, 3]. BI-RADS-MRI determines malignancy risk by evaluating lesion morphology and dynamic contrast-enhancement patterns. Category 3 lesions are considered probably benign and have a $\leq 2\%$ likelihood of malignancy; category 4 lesions are considered suspicious and have a $> 2\%$ but $< 95\%$ likelihood of malignancy; and category 5 lesions are highly suggestive of malignancy and have a $\geq 95\%$ likelihood of malignancy [2]. A previous meta-analysis reported a malignancy rate for category 3 mass lesions of 1.5% [30]. A multi-center prospective study found a malignancy rate for category 5 lesions of 71.4% [31]. This rate is lower than 95%, but mass lesion and non-mass enhancement was not distinguished in this study. Another single-center study with a relatively large number of lesions reported a malignancy rate for category 5 mass lesions of 99.2% [32]. Thus, categories 3 and 5 represent very high probabilities of benignity and malignancy, respectively, whereas category 4 spans a broad intermediate range. In clinical practice, BI-RADS category 4 or 5 lesions require histopathological confirmation via invasive biopsy (e.g., needle biopsy, mammotome) or surgery. Therefore, invasive tissue diagnosis is performed on category 4 lesions in clinical practice, effectively functioning as a “BI-RADS-MRI positive” result. BI-RADS-MRI is thought to be designed to have high sensitivity so as not to miss malignant lesions. This has also led to a high false-positive rate of BI-RADS-MRI in our study.

BI-RADS-MRI categorization demonstrates high sensitivity but variable specificity for breast cancer [4]. The variable specificity of BI-RADS-MRI may lead to unnecessary biopsy. In this study, BI-RADS-MRI exhibited high sensitivity but low specificity in the test dataset, consistent with previous reports [4]. New imaging approaches are needed to reduce unnecessary biopsies in breast cancer diagnosis.

Previous research suggests that incorporating manually measured quantitative values from synthetic MRI alongside DCE-MRI evaluation improves diagnostic performance for BI-RADS-MRI category 4 lesions [33]. In our study, the machine learning model utilizing radiomics features from synthetic MRI's pre- and post-contrast quantitative value maps achieved relatively high specificity in the test dataset. We developed an ensemble algorithm combining BI-RADS-MRI categorization with a machine learning model using radiomics features from synthetic MRI. The ensemble algorithm's AUC was significantly higher than that of BI-RADS-MRI and individual machine learning models. The machine learning model reclassified BI-RADS-MRI category 4 lesions as benign or malignant, improving specificity. Our findings indicate that diagnostic performance improves when a

high-specificity machine learning model is applied to BI-RADS-MRI category 4 lesions.

The ensemble algorithm combining BI-RADS-MRI and machine learning with radiomics features from weighted images achieves high diagnostic performance by reclassifying BI-RADS-MRI category 4 lesions [34, 35]. Hao et al. reported that an SVM incorporating radiomics features from relative images achieved an AUC of 0.77 in diagnosing contralateral BI-RADS-MRI category 4 lesions in patients with breast cancer [34]. Similarly, Cui et al. demonstrated that an SVM with radiomics features from relative images attained an AUC of 0.919 in distinguishing benign from malignant BI-RADS-MRI category 4 lesions [35]. To our knowledge, this is the first study evaluating an ensemble algorithm that integrates BI-RADS-MRI with machine learning using radiomics features derived from quantitative value maps.

In this study, we reduced the number of features using the VIF and the Boruta algorithm. Among the retained features, those identified as important in the LR model included menopausal status, minor axis length, age, and select TEMs. The finding that age is a key feature aligns with statistical data indicating that breast cancer incidence varies by age [1]. The significant texture features include heterogeneity indicators such as zone entropy; however, their interpretation remains challenging as they represent texture feature ratios. Further research is required to compare texture feature ratios with pathological benign and malignant outcomes to better understand their clinical significance. Minor axis length likely reflects the observation that larger lesions have a higher probability of malignancy.

This study has some limitations. First, it is a single-institution, single-vendor observational study. Previous research in head MRI demonstrated high reproducibility of quantitative values using synthetic MRI across different MRI vendors [36]. However, evidence regarding the reproducibility of synthetic MRI-derived quantitative values in breast imaging is limited. Further research using multiple vendors and institutions is needed to assess reproducibility in breast MRI. Second, selection bias may be present. This study retrospectively analyzed enhanced breast mass lesions with histopathological diagnoses. These lesions were selected based on clinical indications of biopsy following detailed evaluation with breast MRI after screening tests such as ultrasound and mammography, leading to a higher malignancy prevalence. In routine clinical practice, not all breast MRI cases undergo histopathological confirmation. Although benign lesions were oversampled using ADASYN in the cross-validation dataset, the original dataset may still have limited benign lesion diversity. Third, non-mass enhancement (NME) lesions on DCE-MRI were not included in this study. Notably, ductal carcinoma

in situ, a common malignancy, frequently presents as NME lesions [37]. Further research is needed to evaluate NME lesions in future studies. Fourth, this method requires manual segmentation along mass contours. In this study, DCE-MRI segmentation was performed manually by radiologists. To enable end-to-end automation in clinical practice, developing a robust algorithm for lesion segmentation is necessary. Fifth, feature interpretability remains limited in this method. As this study relied on retrospective histopathological data for benign and malignant tumors, evaluating the correlations between texture feature ratios and pathological lesion characteristics was challenging. Prospective studies are necessary to investigate these features alongside comprehensive histopathological data. Sixth, the model proposed in this study is based on a small dataset and traditional machine learning algorithms. Since we collected clinical cases from a single center, the dataset is relatively small for machine learning. In addition, we used traditional machine learning methods and did not verify innovative methods. To build a higher-performance machine learning model, it is necessary to collect a large dataset and develop more innovative models.

Conclusions

We evaluated machine learning models utilizing radiomics features from synthetic MRI-derived quantitative value maps for differentiating benign and malignant breast masses. Our findings suggest that diagnostic performance improves when an ensemble algorithm integrating BI-RADS-MRI and machine learning is applied.

Abbreviations

BI-RADS	Breast Imaging Reporting And Data System
MRI	Magnetic Resonance Imaging
MDME	Multi-Dynamic Multi-Echo
AUC	Area Under the Curve
DCE-MRI	Dynamic Contrast-Enhanced Magnetic resonance Imaging
NAC	Neoadjuvant chemotherapy
TNBC	Triple-Negative Breast Cancer
AI	Artificial Intelligence
TR	Repetition Time
TE	Echo Time
FOV	Field Of View
DWI	Diffusion-Weighted Imaging
RFS	Radiomics Feature Space
TEM	Textural Evaluation Metric
VIF	Variance Inflation Factor
RF	Random Forest
XGB	eXtreme Gradient Boosting
GB	Gradient Boosting
LGB	Light gradient Boosting
LR	Logistic Regression
KNN	K-Nearest Neighbor
CART	Classification And Regression Tree
SVM	Support Vector Machine
ADASYN	Adaptive Synthetic Sampling Approach for Imbalanced Learning
ROC	Receiver Operating Characteristic
SHAP	SHapley Additive exPlanations

Supplementary Information

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Supplementary Material 1

Supplementary Material 2

Supplementary Material 3

Supplementary Material 4

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Author contributions

T. M., M. M., H. H., and S. F. contributed to the study conception and design. Material preparation, data collection, and analysis were performed by T. M., M. M., H. H., S. F., M. M., Y. S., W. T., K. O., M. K., Y. K., and R. K. The first draft of the manuscript was written by T. M., and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability

The dataset used and analyzed during the current study is not publicly available due to sensitivity and ethical reasons but is available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This retrospective study involving human participants was also in accordance with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. This study also complies with the "Ethical Guidelines for Medical and Health Research Involving Human Subjects" by Japanese Ministry of Health, Labour, and Welfare (MHLW). The study was approved by the Institutional Review Board (IRB) of Ehime University Hospital (Approval No: 2104006) and was performed in line with the principles of the Declaration of Helsinki. The study information was described on the hospital and laboratory website, and participants were provided with the opportunity to opt out. Informed consent was not obtained from the enrolled patients because it was waived by the IRB of Ehime University Hospital and was deemed unnecessary according to the "Ethical Guidelines for Medical and Health Research Involving Human Subjects" by Japanese MHLW.

Consent for publication

Not applicable.

Competing interests

Hasnine Haque and Shotaro Fuchibe are employees of GE HealthCare Japan.

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